Lifeline: conversation record (28 Oct 2026)

Cynefian CC-02, 24-Hour Student Hackathon, Cloud Computing track. Problem: keep a 4-hospital, 900-bed group running when its single data centre fails, with a 3-person IT team, no DR site and a limited budget.

1. Problem understood

RTO = how long a system can be down. RPO = how much data can be lost. Tiering: pharmacy, lab and patient records first (Tier 1); radiology images (Tier 2); billing (Tier 3); HR and file shares (Tier 4). Deliverables: tier table, architecture, live restore demo, runbook, monitoring and alerts, monthly cost. Rules: data stays in India, encrypted, synthetic data only, role-limited and logged access, aligned with DPDP Act 2023 and ABDM.

2. What was built

Browser console with tiers, live drill, backup health, cost and runbook tabs; a launch-style website; a Node.js backend (lifeline-dr.zip) with encrypted backups, checksums, fallback, roles and a hash-chained audit log; then a simple one-screen app.

3. Final app

Live link: https://claude.ai/artifact/H7Yafs3SDTjrowhK3GYvtv . One DISASTER button: backs up all systems (AES-256-GCM, SHA-256, two copies), destroys them, corrupts one local copy on purpose, restores in tier order, and checks the restored data matches. Example data for every system can be edited, added to, deleted, or reset; edits are saved on the device. Standalone file: lifeline-app.html.

4. Runbook

1 Declare the incident and start the clock. 2 Check backup health and pick the newest valid copy. 3 Fail over Tier 1 to warm standby. 4 Restore Tier 2 from the local copy. 5 Restore Tiers 3 and 4 from offsite. 6 Verify row counts and checksums. 7 Record measured time and data loss. 8 Review and fix gaps.

5. What to tell judges

Encryption, checksums, restore and timing are real and measured in the browser. Large-system recovery time is a projection. Cost rates are assumptions to replace with a real quote. Data is synthetic only. A production version needs a backend and an India-region bucket.